

(EST MINI PROJECT)

AI FOR A GREENER FUTURE

AI Project Report

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ABSTRACT

Air pollution is a major environmental concern that adversely affects human health, ecosystems, and overall quality of life. Accurate prediction of air pollution levels can help in understanding pollution patterns and supporting better environmental planning. This project, **Air Pollution Prediction Using Linear Regression**, aims to predict the concentration of Respirable Suspended Particulate Matter (RSPM) using historical air pollution data and machine learning techniques.

The project uses a dataset containing air pollution records, which is processed to handle missing or incomplete values and prepare the data for model development. Linear Regression, a supervised machine learning algorithm, is employed to establish a relationship between the available input features and the target variable, RSPM. The trained model is intended to estimate pollutant concentrations based on the patterns identified in the dataset. Model performance can be evaluated using appropriate regression metrics, such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and the coefficient of determination ((R^2)).

The proposed system demonstrates how machine learning can be applied to environmental data analysis and pollution prediction. It provides a foundation for identifying pollution trends and supporting data-driven environmental decision-making. However, the reliability of predictions depends on the quality, completeness, and relevance of the available data. The project highlights the potential of Linear Regression as a simple and interpretable approach to air pollution prediction and suggests opportunities for future improvement through additional environmental parameters and more advanced machine learning models.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to Dr. Jyoti for her valuable guidance, encouragement, and support throughout our project, **Air Pollution Prediction Using Linear Regression**.

Her clear explanation of the machine learning workflow, from defining the problem and preparing the data to training and evaluating a model, helped us understand that a project involves much more than simply writing code. Her emphasis on justifying every decision, from data preprocessing to algorithm selection and evaluation metrics, shaped our approach at each stage. Her valuable feedback and suggestions helped us improve our analysis and presentation of the project.

We are also thankful to Thapar Institute of Engineering & Technology for providing us with the opportunity and resources to carry out this project. We extend our gratitude to the contributors of the air pollution dataset used in this project and to the developers of the open-source Python libraries that made this work possible.

Finally, we would like to thank our families and friends for their constant support, encouragement, and motivation throughout the completion of this project.

DECLARATION

We hereby declare that the project entitled Air Pollution prediction using linear segregation is an authentic record of our own work carried out in the Electronics and Computer Science Engineering Department, TIET, Patiala, under the guidance of Dr.Jyoti.

Date:19 October 2026

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LIST OF ABBREVIATIONS

ML	**Machine Learning**
EDA	**Exploratory Data Analysis**
RSPM	**Respirable Suspended Particulate Matter**
SPM	**Suspended Particulate Matter**
PM2.5	**Particulate Matter smaller than 2.5 micrometres**
SO2	**Sulphur Dioxide**
NO2	**Nitrogen Dioxide**
NaN	**Not a Number (a missing value)**
MAE	**Mean Absolute Error**
MSE	**Mean Squared Error**
RMSE	**Root Mean Squared Error**
R^2	**Coefficient of Determination**

1.1 INTRODUCTION

If you have ever looked out of a window in North India in winter and wondered why the air feels heavy, you have already met the problem this project tries to tackle. Fine dust particles, known as **RSPM** (Respirable Suspended Particulate Matter), get deep into our lungs and are one of the biggest air-quality worries in Indian cities.

Measuring RSPM needs special equipment, and monitoring stations do not always record every pollutant. So the question we asked was simple: **can we estimate RSPM using the other things that stations measure, such as sulphur dioxide (SO2), nitrogen dioxide (NO2), larger dust (SPM), plus where and when the reading was taken?**

The answer, as the rest of this report shows, is "partly yes". It is a good first step, and it points clearly to what to try next.

1.2 Dataset Description

We used a sample of **50,000 air-quality records** collected across India from 2003 onwards, with 13 columns per record.

What it tells us	Columns
Where the reading was taken	state, location, type of area (industrial, residential and so on)
When	sampling date, which we turned into year and month
What was measured	SO2, NO2, RSPM, SPM, PM2.5
Admin details	station code, agency, monitoring station name

In plain numbers: across the dataset, the average RSPM reading was about **109**, the middle (median) reading was **90**, and the highest was an extreme **6,307**, which is almost certainly a recording error and a good reminder that real-world data is messy.

DATASET LINK

1 https://www.kaggle.com/datasets/divanshi0414/air-quality-dataset

1.3 Data Cleaning and Preprocessing

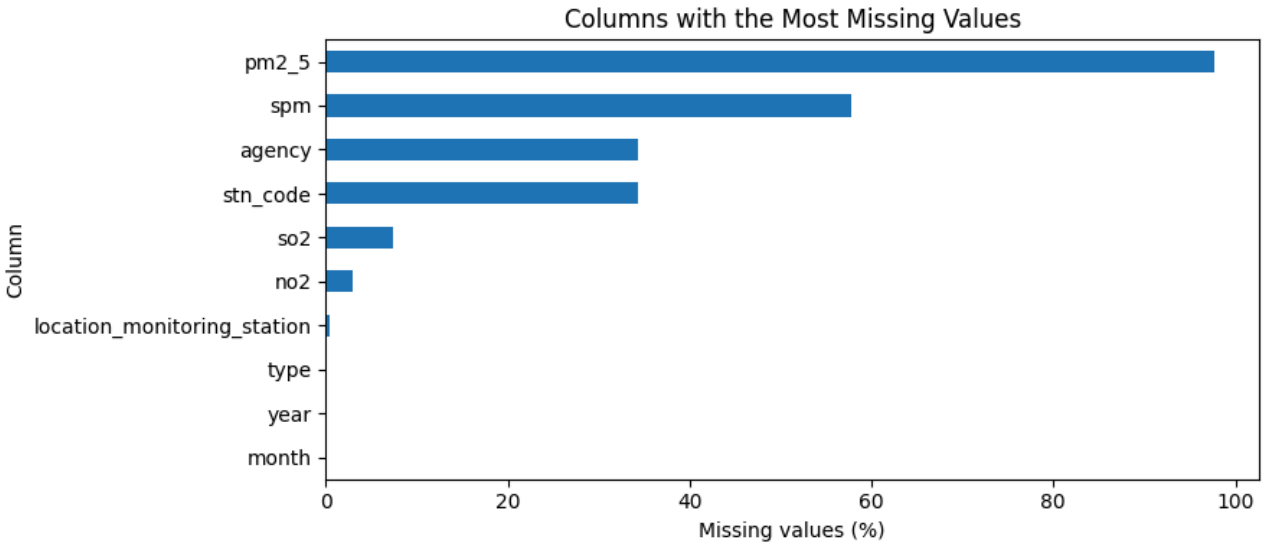
Good models start with honest data. Here is what we found and did:

- **Lots of gaps.** PM2.5 was missing in about **97.7%** of records and SPM in about **57.8%**. Station code and agency were missing in about 34%.
- **Duplicates.** 10 repeated rows were removed.
- **No target, no training.** RSPM is the value we want to predict, so rows without it cannot teach the model anything. After cleaning, **49,990 records** remained.
- **Dates into features.** Dates were converted into proper date values, and then split into **year** and **month**, so the model can pick up on trends over the years and on seasons.
- **Dropping the noise.** Columns that only identify a record, such as station code, agency and monitoring-station name, do not help predict dust levels, so they were left out.

The final inputs to the model were: **state, location, area type, SO2, NO2, SPM, PM2.5, year and month**.

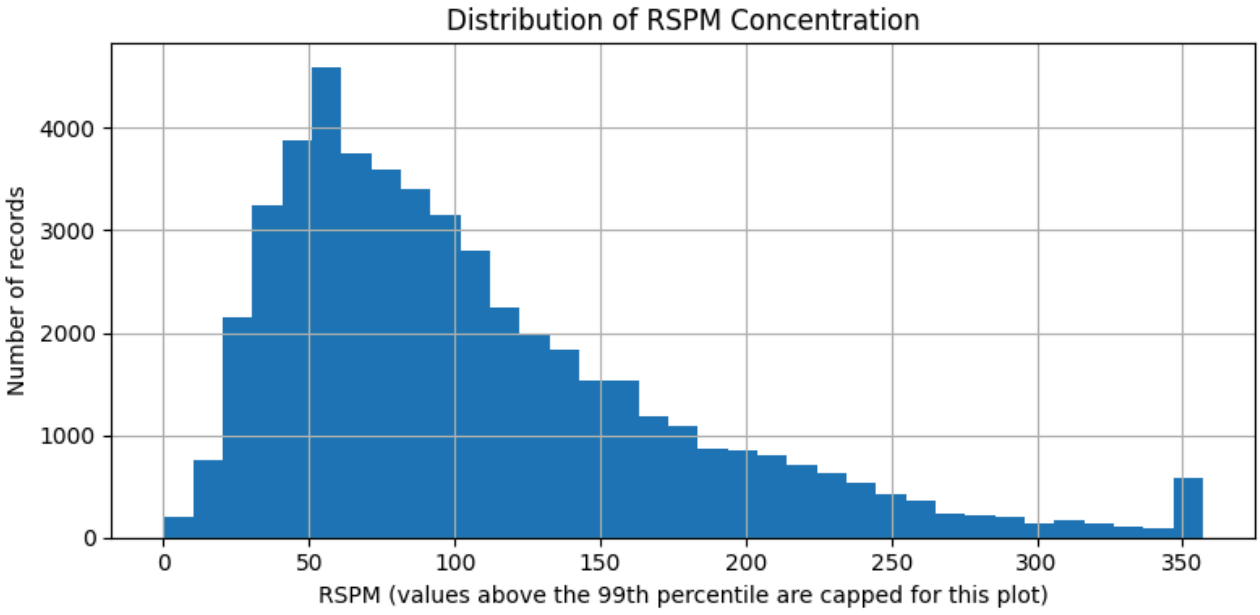
1.4 Exploratory Data Analysis

Figure 1: Where the data is incomplete



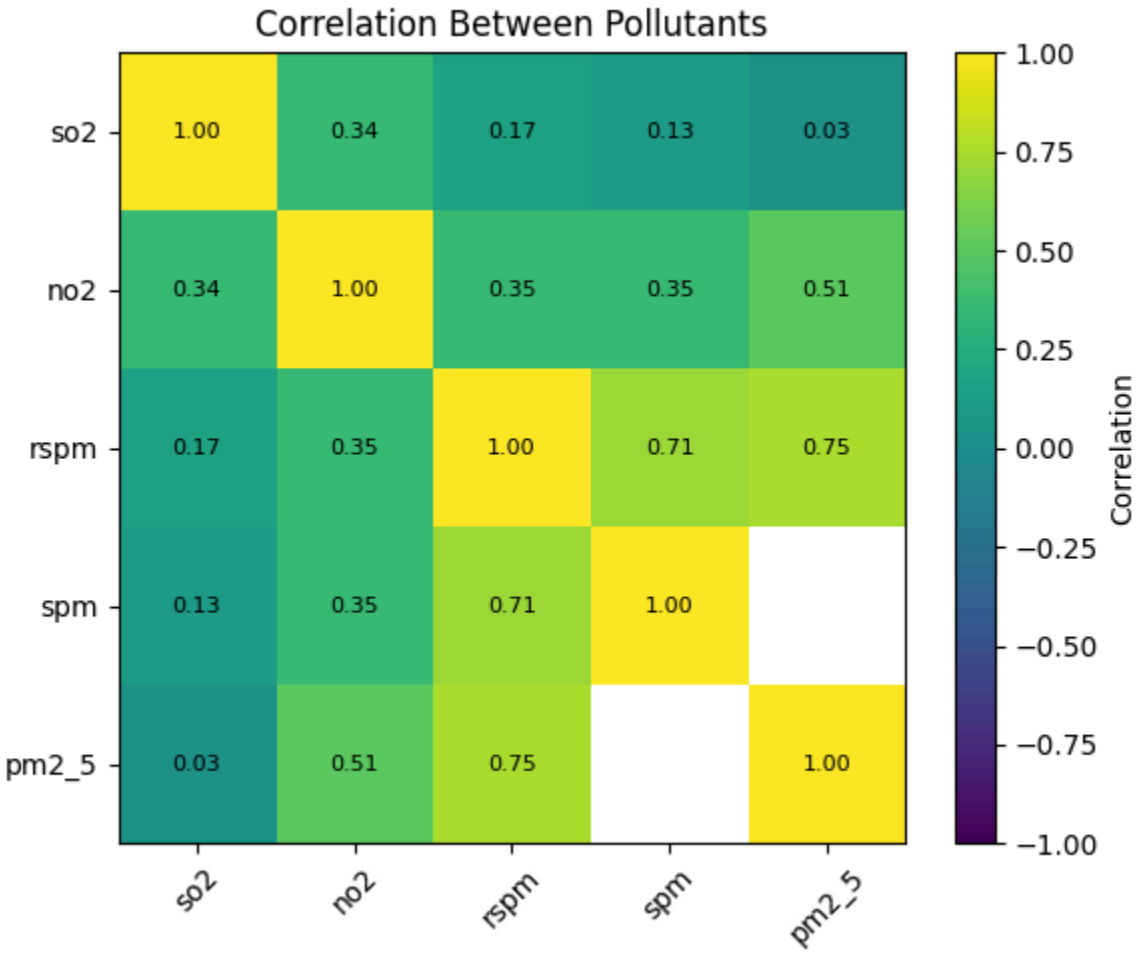
The longest bars show PM2.5 and SPM, which are mostly empty. This tells us we should be careful about relying on them, and it explains why we had to fill gaps sensibly before training.

Figure 2: How RSPM readings are spread



Most readings sit at low-to-moderate levels, with a long tail of high-pollution days. The chart is capped at the 99th percentile so that a handful of extreme values do not squash everything else into a thin sliver.

Figure 3: Which pollutants move together



This heatmap is the most useful picture for understanding the model. Reading it:

- RSPM and PM2.5 (0.75) and RSPM and SPM (0.71) are strongly related. That makes sense, since they are all kinds of airborne dust, so SPM and PM2.5 are valuable clues for predicting RSPM.
- RSPM and NO2 (0.35) have a moderate link, often tied to traffic and industry.
- RSPM and SO2 (0.17) are only weakly related.

- The blank square between SPM and PM2.5 means there were not enough records with both measured to calculate a value.

1.5 Model Building

The recipe in plain English

1. Split the data 80/20. 39,992 records were used for learning, and 9,998 were hidden away for an honest final exam. A fixed random seed (42) means anyone can repeat the exact same split.
2. Fill the blanks. Missing numbers were filled with the median, which is safer than the average when extreme values exist. Missing text values (like area type) were filled with the most common one.
3. Turn words into numbers. Computers cannot do maths on "Punjab" or "Ludhiana", so each state, location and area type was converted into 0/1 columns using one-hot encoding.
4. Train Linear Regression. The model finds the best straight-line combination of all the inputs to predict RSPM.
5. Bundle it all in a Pipeline. Steps 2 to 4 are chained together, so the exact same cleaning is applied to new data automatically, which prevents mistakes and data leakage.

Why Linear Regression?

It is simple, fast and easy to explain. It is the sensible baseline: a first result that any fancier model later has to beat.

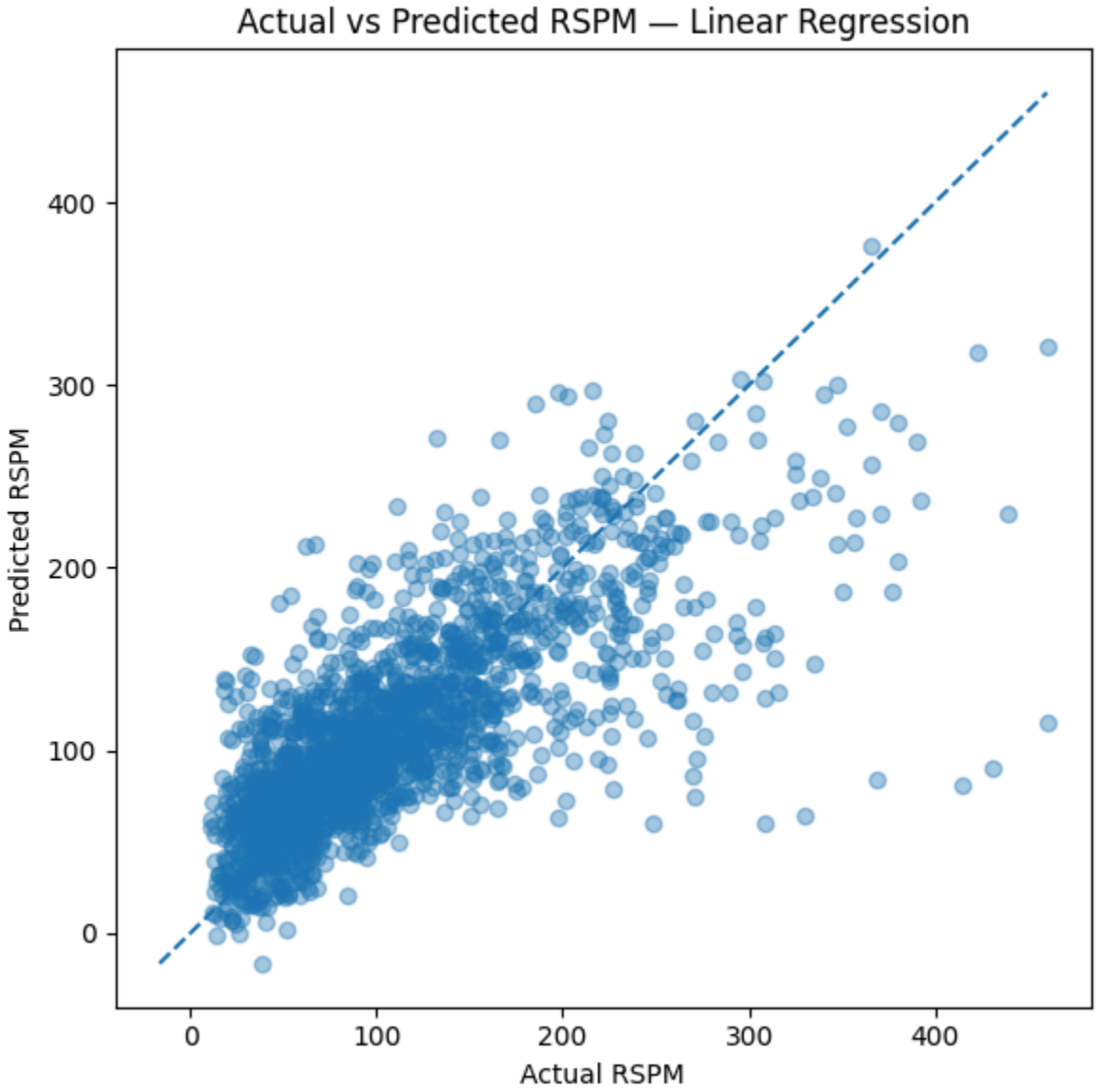
1.6 Results and Evaluation

The model was tested on the 9,998 records it had never seen.

Measure	Score	What it means in everyday terms
MAE	32.88	On average, predictions are off by about 33 units of RSPM
RMSE	49.40	Similar idea, but big misses count extra, so it is higher
MSE	2440.31	The squared version of the error (used mainly for comparing models)
R^2	0.564	The model explains about **56%** of why RSPM goes up and down

A sample of the model's predictions shows both its strengths and weaknesses. For example, it predicted 69.7 when the true value was 72, and 192.8 when the true value was 197, which are very close. But it predicted 38.6 for a true value of 72, and 248.9 for a true value of 338, which miss by a lot.

Figure 4: Actual vs predicted RSPM



Each dot is one test record. If the model were perfect, every dot would sit on the dashed diagonal line. Here is what we see:

- **Low and medium readings are predicted reasonably well**, as the dense cloud follows the line.
- **High-pollution readings are under-predicted.** The dots in the right half sit below the line, which means the model tends to play down the worst days.
- A few predictions are negative, which is impossible for a dust concentration. This is a known quirk of straight-line models.

1.7 Strengths of the Project

- A complete, repeatable pipeline from raw file to tested predictions.
- Careful handling of missing data instead of throwing most of the dataset away.
- Fair evaluation on unseen data, with four different measures.
- Clear visuals that explain the data, not just the model.

1.8 Limitations

- **A moderate score.** R^2 of 0.564 is a decent baseline but not a final product.
- **Extreme values.** A few absurd readings (up to 6,307) can pull a linear model off course.
- **Mostly-empty inputs.** PM2.5 is filled in for fewer than 3% of rows, so most of its values are the median, which adds little real information.
- **Zeros that may not be real.** Some early rows show 0 for every pollutant at once, which looks more like missing data than clean air.
- **Straight-line assumption.** Air pollution depends on seasons, weather and geography in curved, interacting ways that Linear Regression cannot capture.

1.9 Future Scope

1. **Try stronger models**, such as Random Forest or Gradient Boosting, which handle curves and interactions well.
2. **Treat outliers and fake zeros** before training.
3. **Add weather data** such as temperature, humidity and wind, which strongly affect dust levels.
4. **Predict in log scale** to stop extreme values from dominating, and prevent negative predictions.
5. **Use cross-validation** for a more trustworthy score than a single split.
6. **Add a seasonal feature**, since winter in North India behaves very differently from monsoon.

1.10 Tools and References

Python, Google Colab, pandas and NumPy (data handling), Matplotlib (charts), and scikit-learn (SimpleImputer, OneHotEncoder, ColumnTransformer, Pipeline, LinearRegression, train_test_split and the evaluation metrics).

1. Kaggle, Loan Prediction dataset (loan.csv).

2. Pedregosa et al., scikit-learn: Machine Learning in Python, official documentation (LogisticRegression, GridSearchCV, StandardScaler, metrics).

3. Andrew Ng, Machine Learning Specialization, Course 1 (logistic regression, cost function, regularization).

1.11 Conclusion

This project set out to answer a practical question: can the level of fine dust (RSPM) in India's air be estimated from the other readings that monitoring stations record? Starting from 50,000 raw and messy records, we cleaned the data, handled missing values, turned locations and dates into usable features, and trained a Linear Regression model inside a clean, repeatable pipeline.

What we found

- The model explains about 56% of the variation in RSPM ($R^2 = 0.564$), with an average error of roughly 33 units (MAE = 32.88) and an RMSE of 49.40.
- SPM and PM2.5 are the strongest clues for predicting RSPM (correlations of 0.71 and 0.75), while SO2 contributes very little.
- The model is dependable on everyday readings but under-predicts the most polluted days, which are exactly the days that matter most for public health.

What we learned

The biggest lesson was that good data preparation matters as much as the choice of model. Missing values, extreme outliers and suspicious zeros shaped the results far more than any algorithm setting. Building everything as a pipeline also meant the same steps are applied identically to any new data, which keeps the results trustworthy and easy to reproduce.

Where it goes from here

This work is best seen as a strong, honest baseline rather than a finished product. Moving to models such as Random Forest or Gradient Boosting, cleaning the outliers, and adding weather and seasonal information should capture the curved, complex behaviour of air pollution and improve accuracy on high-pollution days.

In short, even a simple model can turn scattered station readings into a useful picture of the air we breathe, and with better data and smarter models, it can grow into a tool that helps people and policymakers act before the air turns dangerou**s.**